

these processors work together, the slowest one creates a bottleneck. If you use your PC for gaming, you would probably have experienced this problem. It does not matter how fast your GPU is if your CPU is slow.

Similarly, it would not matter how fast hardware accelerators are if the read/write data transfer is slow between RAM storage and the processor. Hence, it is crucial for designers to select the right hardware for all components to be in sync.

Popular AI accelerator architectures

Here are some popular AI architectures.

Graphics processing unit (GPU):

Well, GPUs were not originally designed for ML and AI. When AI and deep learning were gaining popularity, GPUs were already in the market and were used as specialised processors for computer graphics. Now we have programmable ones too, called general-purpose GPU (GPGPU). Their ability to handle computer graphics and image processing makes them a good choice for usage as an AI accelerator. In fact, GPU manufacturers are now modifying their architectures for use in AI or ML.

In 2012, Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton from the University of Toronto presented a paper on AlexNet, a deep neural network. It was trained on readily available, programmable consumer GPUs by NVIDIA (NVIDIA GTX 580 3GB GPU). It is basically a GPU implementation of a CNN (convolutional neural network). AlexNet won the 2012 ImageNet Large Scale Visual Recognition Challenge (ILSVRC).

FPGAs and ASICs: FPGAs are semiconductor devices that, as the name suggests, are field programmable. Their internal circuitry is not mapped when you buy them; you have to program them right on the field using hardware description

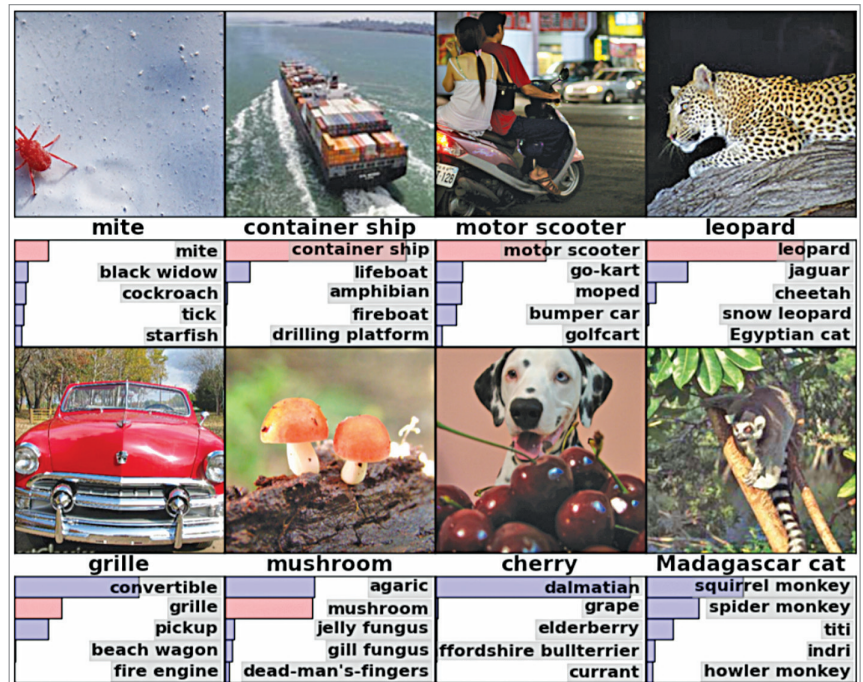


Figure 2: AlexNet's most probable labels on ImageNet images, with five labels considered most probable for each. The probability assigned to each label is also shown by the bars (Credit: Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton)



Figure 3: Xilinx' Alveo U50 data centre accelerator card that comes with the Zebra software (Credit: Xilinx)

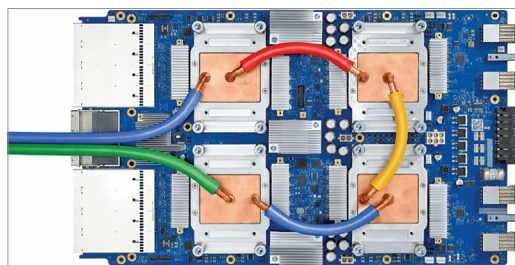


Figure 4: Cloud TPU v3 (Credit: Google)

languages (HDLs) like VHDL and Verilog. This means that your code is converted into a circuit that is specific to your application. The fact that they can be customised gives them a natural advantage over GPUs.

Xilinx and an AI startup called Mipsology are working together to enable FPGAs to replace GPUs in AI accelerator applications, just using a single command. Mipsology's software Zebra converts GPU code to run on Mipsology's AI compute engine on an FPGA, without any code changes.

FPGAs are cheaper, reprogrammable, and use less power to accomplish the same work. However, this comes at the cost of speed. ASICs, on the other hand, can achieve higher speeds and consume low power.

Moreover, if ASICs are to

be manufactured in bulk, the costs are not that high. But they cannot be reprogrammed on the field.

Tensor processing unit (TPU):

Tensor processing unit is an AI accelerator built by Google specifically